



BAITED REMOTE UNDERWATER VIDEO SYSTEMS (BRUVS)

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DATA TYPE:

Species present
Abundance and size

SHARK PLAN OBJECTIVES:

- 6. Contribute to the protection of biodiversity and ecosystem structure and function.
- 10. Facilitate the identification and reporting of species-specific biological and trade data.

WHY WOULD YOU USE THIS TOOL?

BRUVS surveys are a valuable tool for determining which species are present in an area and estimating relative abundance between areas. They're a non-destructive sampling method that's simple to repeat, reliable and cost-effective; and they're particularly useful for large-scale sampling of sharks and rays in a relatively short time.

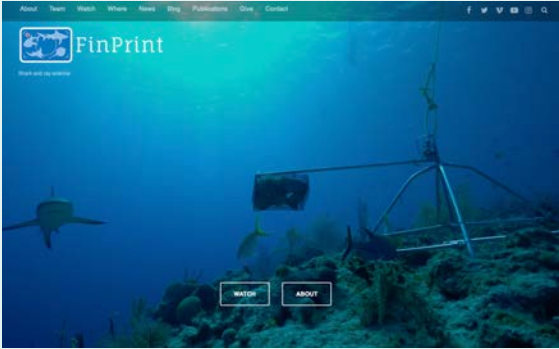
BRUVS sample the sharks and rays in an area without humans in the water, whose presence might alter the behaviour of the animals. This unique characteristic of the sampling method sets it apart from traditional visual sampling by underwater visual census (UVC). BRUVS have minimal impact on the environment.

The BRUVS tool provides a permanent visual record of the species community composition. It can also be used to survey the habitat in the sampling area. Due to their ease of use in a variety of habitats, BRUVS may record the presence of shark and ray species in areas that have not been covered by fishery surveys.

BRUVS can be used to study the effects of fishing and marine protected areas (MPAs) by comparing the diversity and abundance of sharks and rays within an MPA to those in similar nearby habitats exposed to fishing pressure. BRUVS with two cameras (stereo BRUVS) can be used to accurately measure the size of sharks and rays. Length data is useful for estimates of maturity stage, which gives an insight into the population structure of the species in the area.



BRUVS
PROVIDES A
PERMANENT VISUAL
RECORD OF THE
SPECIES COMMUNITY



GLOBAL FINPRINT

The Global FinPrint project documents shark and ray populations in tropical coral reef ecosystems across the world. It provides important baseline data on species diversity, abundance and distribution across a large number of countries and reefs.

BRUVS data can be used to compare reefs with different features to determine the factors (e.g. habitat type, fishing pressure, prey fish densities) that influence shark and ray diversity and abundance. Global FinPrint has collected valuable information for shark and ray conservation and management, and it can be made available for local use.

ADVANTAGES:

- Cheap to build and use
- Collect large amounts of data
- Stereo BRUVS can measure lengths of individuals

LIMITATIONS:

- Results are biased to species attracted to bait, which may mean other species are underestimated
- Video does not work well in murky waters
- Footage analysis is labour-intensive
- Stereo BRUVS require calibration and potentially expensive software

WHAT ARE BRUVS?

BRUVS units combine bait to attract sharks and rays and an underwater video camera to record what animals come to feed on the bait. The units themselves consist of a frame with housing to hold a camera. A bait arm is attached to the frame, which includes a bait bag filled with crushed oily fish.

BRUVS are placed on the seafloor with a surface float for retrieval. Their frames are weighted to ensure they don't float or move. BRUVS are typically left on the seafloor for 60-90 minutes and then recovered. Observers can watch the video footage to determine the species and number of individuals filmed.

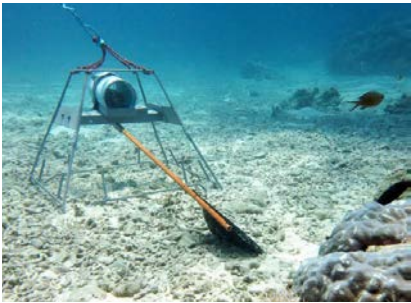


Fig 1: BRUVS on seafloor with bait arm with bait bag. (Source: Michelle Heupel)

Stereo BRUVS are more complicated than single camera BRUVS, with a camera mounted on each side of the bait arm. They also require camera calibration before use, so a calibration cube is needed.



Fig 2: Stereo-BRUVS. (Source Steve Lindfield, UWA, NOAA – Image courtesy of pifsc.noaa.gov)

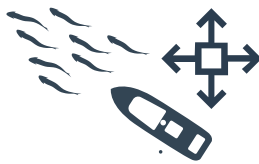
METHOD: BRUVS



DESIGN PROGRAMME



CONSTRUCT BRUVS



DEPLOY BRUVS



VIEW FOOTAGE

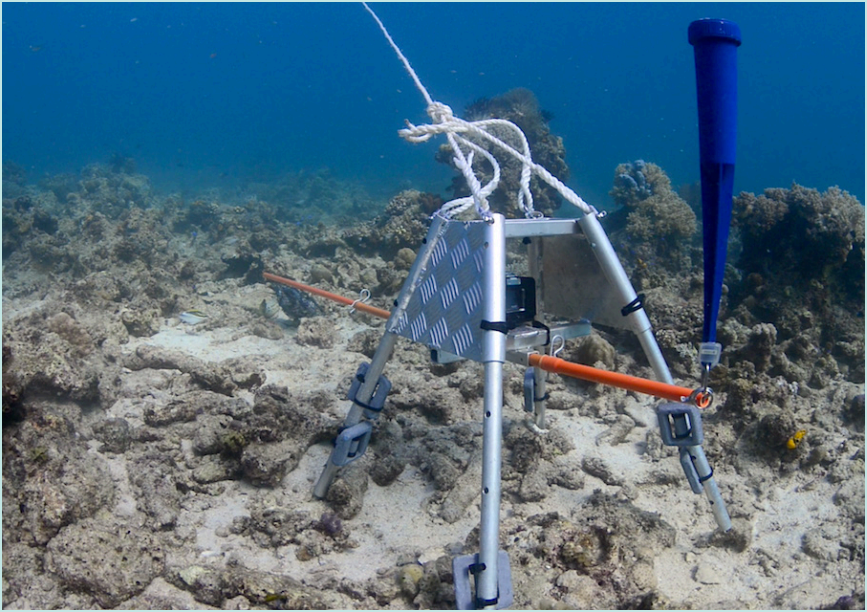


Fig 3: Stainless steel BRUVS frame with detachable legs cabled-tied in place. Dive weights are cable-tied on each leg. Holes have been drilled in the legs to allow the frame to be set at different heights. A current-meter is attached to the back end of the bait arm. (Source: Christian Sloater, Scubazoo)



Fig 4: Acrylic BRUVS with a camera inside each plastic drain pipe. (Source: Vanessa Jaiteh)



DESIGNING THE PROGRAMME

Design the BRUVS sampling programme to address the objectives of the study. Consider the resources available for equipment, vessels, BRUVS deployments and analysis of the footage. This will affect the type of BRUVS used, the number of BRUVS and how many times they can be deployed.

Sampling designs vary: for example, a comparison of species diversity and abundance between areas open and closed to fishing will require a different design to an assessment of species diversity and abundance across all habitats in an area. Testing between fished and unfished areas simply requires the same number of BRUVS deployments in each location, while association of abundance with habitat requires representative numbers of deployments from all available habitat types.

For all sampling designs it's important to consider habitat type, as this can have a strong influence on the species present. Complex coral reef habitats are likely to have different types and numbers of shark and ray species to those present in open sandy areas. The habitat type must be recorded, otherwise it's difficult to determine whether any difference in diversity is due to the habitat or another factor such as the level of fishing pressure.



CONSTRUCTING BRUVS UNITS

A BRUVS unit is made of a frame, bait pole, bait bag and a camera. All units deployed in a survey should be identical in size and height off the bottom: most BRUVS are designed to stand 1m or less above the substrate, and ideally the substrate should take up half the field of view. The bait bag can either be set on the bottom, or raised in the water column.

FRAME

The BRUVS frame can be of any size and style and can be constructed from a variety of easily obtainable materials such as stainless steel, rebar, PVC pipe, acrylic, plastic or timber. Detachable legs make transport easier.

The frame needs to have weights attached to its legs to keep it steady on the seafloor. More weights are needed for lighter frames and in stronger currents – shorter frames are more stable in the latter. Points of attachment are needed for the rope and surface float.

Finally, the frame requires a housing for the camera. This should hold it steady during deployment but include a simple method for attachment and removal.

BAIT BAG, POLE AND BAIT

Bait bags can be made from plastic mesh, metal or PVC pipe with holes. In areas with a lot of larger sharks, the use of metal may prevent the sharks pulling the bait bag off the pole. Bait bags are often cable-tied to the bait pole.

The bait pole can be of any material and usually extends 1-1.5m from the camera.

Any type of oily fish makes good bait, but always use fresh bait for each deployment and ensure it's processed in the same way for all drops in a study (e.g. minced, crushed or chopped to release oil and increase attraction). It's important to plan how much bait will be needed for each field trip, where it will be sourced and how it can be stored on a vessel.

CAMERA

Any type of underwater video camera can be used in BRUVS, but GoPro cameras are particularly popular. The footage needs to be clear enough for viewers to reliably identify the sharks and rays it contains.

Filming in high definition mode of 1080p and 30 frames per second provides good resolution. Recording for 90 minutes at this setting will take up 20-25GB of storage space on a memory card.

If a number of BRUVS are to be deployed at the same time, multiple cameras, memory cards and batteries will be needed, so planning is important. Extra battery packs can allow the same camera to be used multiple times in a single day. If BRUVS are to be deployed on consecutive days, video from the memory cards will need to be downloaded and camera batteries recharged overnight. Multi-port chargers are useful to charge multiple batteries at once (**Fig 6**).

Stereo BRUVS have two cameras mounted at an angle – typically 7-8 degrees – on each side of the bait arm. Because these systems are used to make precise measurements, the cameras must be tightly fitted inside the housing



Fig 5: PVC pipe and caps with holes for bait to disperse attracts a tiger shark (*Galeocerdo cuvier*). (Source: Tonga-watch)



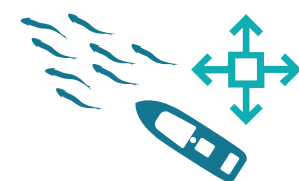
Fig 6: Multi-port charger. (Source: Samm Sherman)

so they can't move. Each stereo BRUVS has its own calibration that relies on maintaining the frame set-up and same camera on the left and right side of the bait arm for each deployment.

Camera technology continues to advance as costs go down. Full-spherical (FS) underwater videos can record a 360° field of view, so it's likely that they'll play a role in many BRUVS in future.



Fig 7: Deploying a BRUVS. (Source: Cordio East Africa-watch)



DEPLOYMENT

BRUVS are typically deployed from a boat. After deployment, it's sensible to keep the boat at least 200m away from the BRUVS to minimize the potential effects of boat noise on shark and ray behaviour.

A standard operating procedure should be followed in each study to minimize the effects of uncontrolled factors. The key factors to standardize are deployment time, amount of bait and depth range.

Deployment times are usually from 60 to 90 minutes on the seafloor. Each deployment typically uses about 1kg of bait. Depth ranges vary depending on the research question – deployments deeper than 50m require a heavy frame and a light source, and a winch may be needed to recover the unit. The length of rope from the BRUVS to the surface float should be around 1.5 times the depth of deployment. In strong currents this length may need to be increased to ensure the surface float is not pulled under.

If multiple BRUVS are deployed at the same time they need to be set far enough apart to minimize the possibility of sharks and rays swimming between them, resulting in individuals being counted more than once. The distance between deployments will depend on the study objectives – in the Global FinPrint project there was a minimum separation of 500m.



VIEWING VIDEO FOOTAGE

A standard method of recording species and counting individuals should be used when watching the video footage. Methods vary between studies, but the most widely used is to record MaxN



Depending on the camera type used, video files may need to be stitched together prior to viewing and analysis. For example, Go Pro video files are segmented during recording and stored separately on the memory card. There are freely available and/or low cost software programs to stitch video segments into a single file for viewing, but this must be done prior to analysis.

(the maximum number of animals visible at any one time).

- 1 Record the habitat type the BRUVS settled in and estimate the water visibility.
- 2 Run the video, and pause it when a shark or ray is observed.
- 3 Identify the species, and sex if possible. If it's difficult to exactly identify the animal, identify it to the lowest taxonomic level possible – for example record the genus of hammerhead (*Sphyrna*) rather than the species of hammerhead (e.g. *Sphyrna lewini*).

- 4 Record MaxN when sharks or rays are present in the frame. MaxN for a species is continually checked during viewing, but it only increases if more individuals of a species become visible in a single frame. The highest MaxN for each species from each video is used for analysis. MaxN is used to assess the relative abundance of species and can be compared among deployments, habitat types and locations. This is a conservative estimate of abundance because more individuals may be present in the area but they won't be counted if they're not in the field of view at the same time.

Due to the need to measure individuals, analysis of stereo BRUVS requires purpose-built software. A commonly used option is EventMeasure (www.seagis.com.au).

- 5 Make backup copies of each BRUVS video in case something happens to the original video footage. Portable hard drives are a good option for this.



Fig 8: Spotted eagle ray (*Aetobatus ocellatus*) feeding. (Source: FinPrint)

LABELS

Plan an effective labelling system before the start of the trip. This is likely to include initials for the trip and location, the date and consecutive numbers for each BRUVS deployment (see [Surveys – labels](#)). Film the site label at the beginning of each BRUVS drop using a slate information board (**Fig 9**).

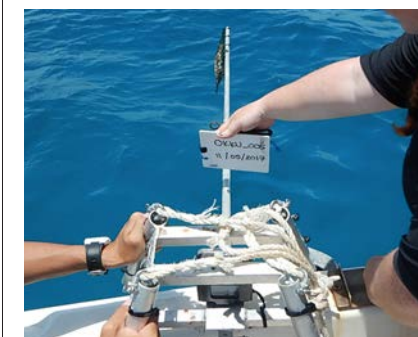


Fig 9: Site label filmed on a slate board at beginning of BRUVS drop. (Source: Samm Sherman)

DATA SHEETS

Ensure data sheets are ready for use in the field. Standard BRUVS inclusions are:

- Date
- Location (place name and latitude/longitude if possible)
- BRUVS label (also filmed at beginning of a BRUVS drop)
- Depth
- Time in (when BRUVS lands on seafloor)
- Time out (when BRUVS leaves seafloor)
- State of tide
- Habitat
- Bait type
- Comments



EQUIPMENT CHECKLIST:

- BRUVS frames, bait arms, bait bags, ropes and surface floats
- Equipment to assemble BRUVS (if required)
- Underwater video cameras, camera batteries, memory cards and chargers
- Camera housings
- Bait
- Weights
- GPS to record drop location
- Depth sounder (or other method to measure deployment depth)
- Waterproof data sheets and clipboard
- Waterproof labels
- Slate board
- Video file-stitching software

Stereo BRUVS – in addition

- Modified BRUVS frame
- Paired video cameras, batteries, memory cards
- Calibration cube
- Stereo BRUVS software for size analysis



TECHNICAL LEVEL – EASY TO MEDIUM

BRUVS surveys are usually done as a team, including someone to drive the boat, someone to deploy and recover the BRUVS, someone to record data etc. Team leaders should have some experience with BRUVS surveys – if not, then training is recommended in BRUVS sampling design, deployment and analysis of footage. Video readers require good knowledge of shark and ray species, or access to a reliable identification guide.



COST – VARIABLE

The cost of BRUVS surveys will vary considerably depending on the location and scale of the project. The major costs are likely to be in travel costs to the study site, vessel time and personnel. The cost of the BRUVS units themselves will depend on the materials used.

Good quality high resolution video can now be gained from relatively inexpensive cameras. However, analysing videos is labour-intensive and can take considerable time depending on the number of deployments and the number of sharks and rays observed on the videos. Purpose-built software such as EventMeasure can assist with video data recording (and it's essential for stereo BRUVS calibration and measurements) but at approximately AU\$ 3,500 for video analysis software and AU\$ 2,000 for calibration software it's a significant investment.

